Safety Data Sheet

REFCOTEC
Where mold and metal meet

Prior Issue Date: 3 June, 2011 **Revision Date:** 19 March, 2015 **Version** 2

1. IDENTIFICATION

Product Identifier

Product Name REFCOSLIK 2500

Other means of identification

SDS Name: SDSL2500
Product Code SL2500

Recommended use of this product:
Foundry Core and Mold Mudding Compound

Supplier Details
REFCOTEC
542 Collins Blvd
Orrville, OH 44667

Emergency Telephone Number

Company Phone Number Information Calls: 330-683-2200
Fax Number: 330-683-8200

Emergency Telephone (24 hr) INFOTRAC 1-352-323-3500 (International)
1-800-535-5053 (North America)

2. HAZARDS IDENTIFICATION

Appearance Black paste **Physical State** Paste **Odor** Mild kerosene odor

Classification

This chemical does not meet the hazardous criteria set forth by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200). However, this Safety Data Sheet (SDS) contains valuable information critical to the safe handling and proper use of this product. This SDS should be retained and available for employees and other users of this product.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	CAS No	Weight-%
Talc	14807-96-6	45-50
Graphite	7782-42-5	15-20
Kerosine, petroleum, hydrodesulfurized	64742-81-0	6 Max
Crystalline silica	14808-60-7	<5

If Chemical Name/CAS No is "proprietary" and/or Weight-% is listed as a range, the specific chemical identity and/or percentage of composition has been withheld as a trade secret.

First Aid Measures

Eye Contact Rinse with plenty of water.

Skin Contact If skin irritation develops, wash product off with soap and water. Remove contaminated clothing.

Inhalation Remove to fresh air.

Ingestion Rinse mouth. Get medical attention.

Most important symptoms and effects

Symptoms May cause mild skin and eye irritation. Overexposure may cause irritation, redness, or itching of skin or eyes and coughing or shortness of breath.

Indication of any immediate medical attention and special treatment needed

Notes to Physician Pre-existing respiratory or dermal conditions may be aggravated by overexposure.

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media Not determined.

Specific Hazards Arising from the Chemical
Unvented containers may rupture or explode when exposed to high heat due to pressure build-up. Move containers to cool area or keep cool with water stream.

Hazardous Combustion Products Carbon oxides.

Protective equipment and precautions for firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Personal Precautions Use personal protective equipment as required.

Environmental Precautions Prevent from entering into soil, ditches, sewers, waterways and/or groundwater. See Section 12, Ecological Information. See Section 13: DISPOSAL CONSIDERATIONS.

Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Clean-Up Sweep up, absorb, or vacuum spill. Do not flush spills to watercourse.

7. HANDLING AND STORAGE

Precautions for safe handling

Advice on Safe Handling

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Use personal protection recommended in Section 8. Do not reuse this container. Send to approved reconditioner for recycling.

Conditions for safe storage, including any incompatibilities

Storage Conditions Keep container tightly closed and store in a cool, dry and well-ventilated place. Keep container closed when not in use.

Incompatible Materials Substances that are incompatible with water.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines

Chemical Name	ACGIH TLV	OSHA PEL	NIOSH IDLH
Talc 14807-96-6	TWA: 2 mg/m^3 particulate matter containing no asbestos and <1% crystalline silica, respirable fraction	(vacated) TWA: 2 mg/m^3 respirable dust <1% Crystalline silica, containing no Asbestos TWA: 20 mppcf if 1% Quartz or more, use Quartz limit	IDLH: 1000 mg/m^3 TWA: 2 mg/m^3 containing no Asbestos and <1% Quartz respirable dust
Graphite 7782-42-5	TWA: 2 mg/m^3 respirable fraction all forms except graphite fibers	TWA: 15 mg/m^3 total dust synthetic TWA: 5 mg/m^3 respirable fraction synthetic (vacated) TWA: 2.5 mg/m^3 respirable dust natural (vacated) TWA: 10 mg/m^3 total dust synthetic (vacated) TWA: 5 mg/m^3 respirable fraction synthetic TWA: 15 mppcf natural	IDLH: 1250 mg/m^3 TWA: 2.5 mg/m^3 natural respirable dust
Kerosine, petroleum, hydrodesulfurized 64742-81-0	TWA: 200 mg/m^3 total hydrocarbon vapor application restricted to conditions in which there are negligible aerosol exposures S*	-	-
Crystalline silica 14808-60-7	TWA: 0.025 mg/m^3 respirable fraction	(vacated) TWA: 0.1 mg/m^3 respirable dust : (30)/(%SiO2 + 2) mg/m^3 TWA total dust : (250)/(%SiO2 + 5) mppcf TWA respirable fraction : (10)/(%SiO2 + 2) mg/m^3 TWA respirable fraction	IDLH: 50 mg/m^3 respirable dust TWA: 0.05 mg/m^3 respirable dust

Appropriate engineering controls

Engineering Controls Ensure adequate ventilation, especially in confined areas. Local exhaust ventilation recommended. Mechanical exhaust if required to keep exposure below recommended limits. Eyewash stations. Showers.

Individual protection measures, such as personal protective equipment

Eye/Face Protection Chemical splash goggles.

Skin and Body Protection Do not use nitrile rubber, polyvinyl alcohol, or polyvinyl chloride. Aprons.

Respiratory Protection Use a NIOSH approved particulate respirator when the potential to exceed established exposure limits exists.

General Hygiene Considerations Handle in accordance with good industrial hygiene and safety practice.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Paste		
Appearance	Black paste	**Odor**	Mild kerosene odor
Color	Black	**Odor Threshold**	Not determined

Property	Values	Remarks • Method
pH	Not determined	
Melting Point/Freezing Point	Not determined	
Boiling Point/Boiling Range	100 °C / 212 °F	
Flash Point	Not Applicable (not a flammable material)	
Evaporation Rate	Not determined	
Flammability (Solid, Gas)	Not Applicable	
Upper Flammability Limits	Not Applicable	
Lower Flammability Limit	Not Applicable	
Vapor Pressure	Less than 10	
Vapor Density	Heavier than air	
Specific Gravity	1.50-1.60	
Water Solubility	Liquid portion partially miscible	
Solubility in other solvents	Not determined	
Partition Coefficient	Not determined	
Auto-ignition Temperature	Not determined	
Decomposition Temperature	Not determined	
Kinematic Viscosity	Not determined	
Dynamic Viscosity	Not determined	
Explosive Properties	Not determined	
Oxidizing Properties	Not determined	
VOC Content	This product contains 4-6% by weight (as kerosene).	

10. STABILITY AND REACTIVITY

Reactivity
Not reactive under normal conditions.

Chemical Stability
Stable under recommended storage conditions.

Possibility of Hazardous Reactions
None under normal processing.

Hazardous Polymerization Hazardous polymerization does not occur.

Conditions to Avoid
Evaporation of the liquid portion will occur if containers are not kept sealed. Keep from freezing.

Incompatible Materials
Substances that are incompatible with water.

Hazardous Decomposition Products
Oxides of carbon.

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure

Product Information

Eye Contact Avoid contact with eyes.

Skin Contact Avoid contact with skin.

Inhalation Avoid breathing vapors or mists.

Ingestion Do not taste or swallow.

Component Information

Chemical Name	Oral LD50	Dermal LD50	Inhalation LC50
Kerosine, petroleum, hydrodesulfurized 64742-81-0	> 5000 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	> 5.2 mg/L (Rat) 4 h
Crystalline silica 14808-60-7	= 500 mg/kg (Rat)	-	-

Information on physical, chemical and toxicological effects

Symptoms Please see section 4 of this SDS for symptoms.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Carcinogenicity Crystalline Silica is considered to be a human carcinogen when in respirable form (dust / powder).

Chemical Name	ACGIH	IARC	NTP	OSHA
Talc 14807-96-6		Group 3		
Kerosine, petroleum, hydrodesulfurized 64742-81-0	A3			
Crystalline silica 14808-60-7	A2	Group 1	Known	X

ACGIH (American Conference of Governmental Industrial Hygienists)
A2 - Suspected Human Carcinogen
IARC (International Agency for Research on Cancer)
Group 1 - Carcinogenic to Humans
Group 3 IARC components are "not classifiable as human carcinogens"
NTP (National Toxicology Program)
Known - Known Carcinogen
OSHA (Occupational Safety and Health Administration of the US Department of Labor)
X - Present

Numerical measures of toxicity
Not determined

12. ECOLOGICAL INFORMATION

Ecotoxicity: The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.

Component Information

Chemical Name	Algae/aquatic plants	Fish	Toxicity to microorganisms	Crustacea
Talc 14807-96-6		100: 96 h Brachydanio rerio g/L LC50 semi-static		
Kerosine, petroleum, hydrodesulfurized 64742-81-0		45: 96 h Pimephales promelas mg/L LC50 flow-through 1740: 96 h Lepomis macrochirus mg/L LC50 static		4720: 48 h Den-dronereides heteropoda mg/L LC50

Persistence/Degradability
Not determined.
Bioaccumulation
Not determined.
Mobility
Not determined
Other Adverse Effects
Not determined

13. DISPOSAL CONSIDERATIONS

Waste Treatment Methods

Disposal of Wastes Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Disposal should be in accordance with applicable regional, national and local laws and regulations.

14. TRANSPORT INFORMATION

Note Please see current shipping paper for most up to date shipping information, including exemptions and special circumstances.

DOT Not regulated
IATA Not regulated
MDG Not regulated

International Inventories
Not determined

US Federal Regulations

SARA 311/312 Hazard Categories

Acute Health Hazard	Yes
Chronic Health Hazard	Yes

SARA 313
Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

US State Regulations
California Proposition 65

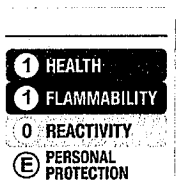
This product contains the following Proposition 65 chemicals.

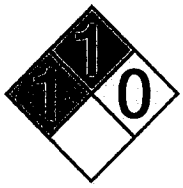
Chemical Name	California Proposition 65
Crystalline silica - 14808-60-7	Carcinogen

U.S. State Right-to-Know Regulations

Chemical Name	New Jersey	Massachusetts	Pennsylvania
Talc 14807-96-6	X	X	X
Graphite 7782-42-5	X	X	X
Crystalline silica 14808-60-7	X	X	X

16. OTHER INFORMATION





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Disclaimer

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End of Safety Data Sheet